



IDEAL GAS

Maxwell-Boltzmann Distribution: (Classical Statistics)

(a) Total probability (P) in the case of N particles distributed over K -cells: -

In an ideal gas, assuming the particles to be identical but distinguishable (i.e. gas molecules) the total probability must involve two factors:

- (i) The "a priori" probability (G) depending on the properties of the cell and
- (ii) the thermodynamic probability (W) which gives the possible sequences of arrangement of particles in the cells. Thus

$$P = GW \quad \text{--- (1)}$$

CALCULATION OF APRIORI PROBABILITY (G):

When 1 particle falls into the i th cell the a priori probability (g_i) is given by

$$g_i = \frac{a_i}{A}$$

where, a_i = area of the i th cell.

& $A = \sum a_i$ = total area of all the cells

Again the a priori probability that 2 particles fall simultaneously into the i th cell will be

$$g_i \cdot g_i = g_i^2$$

As the probability of a composite event is equal to the product of individual probabilities. Thus for n_i particles falling in the i th cell, the a priori probability must be given by: $(g_i)^{n_i}$



Where,

$$\sum_i n_i = (n_1 + n_2 + \dots + n_i) = N$$

= 'Total number of particles.

Hence for the arrangement consisting of

n_1 particles falling in the 1st cell.

n_2 particles falling in the 2nd cell.

n_3 " " " 3rd cell.

n_k particles falling in the k^{th} cell

the a priori probability (G) is given by

$$G = \{ (g_1)^{n_1} (g_2)^{n_2} \dots (g_k)^{n_k} \} \quad \text{--- (2)}$$

Calculation of thermodynamic probability (W)

The no. of ways in which any no. n_1 of particles may be selected from N particles (identical but distinguishable) is given by

$${}^N C_{n_1} = \left\{ \frac{N!}{n_1! (N-n_1)!} \right\}$$

for any n_2 particles selected from remaining $(N-n_1)$ particles is

$${}^{(N-n_1)} C_{n_2} = \left\{ \frac{(N-n_1)!}{n_2! (N-n_1-n_2)!} \right\} \text{ and so on}$$

finally no. of ways is

$$\begin{aligned} & \frac{N!}{n_1! n_2! \dots n_k! (N-n_1-n_2-\dots-n_{k-1})!} \\ &= \left\{ \frac{N!}{n_1! n_2! \dots n_k! (N-n_1-n_2-\dots-n_k)!} \right\} \end{aligned}$$



$$= \left\{ \frac{(N - n_1 - n_2 - \dots - n_{k-1})!}{n_k!} \right\} \quad \left[\because L_0 = 1 \right]$$

$$= \frac{(N - n_1 - n_2 - \dots - n_{k-1})!}{n_k!}$$

Thus,

the total no. of ways (i.e. thermodynamic probability) is given by,

$$W = \left[\left\{ \frac{N!}{n_1! (N - n_1)!} \right\} \left\{ \frac{(N - n_1)!}{n_2! (N - n_1 - n_2)!} \right\} \dots \right. \\ \left. \dots \dots \dots \left\{ \frac{(N - n_1 - n_2 - \dots - n_{k-1})!}{n_k!} \right\} \right]$$

$$\therefore W = \left\{ \frac{N!}{(n_1! n_2! \dots n_k!)} \right\} \quad \text{--- (3)}$$

From eqⁿ (1), (2) and (3) we obtain

$$P = \left[\left\{ (g_1)^{n_1} (g_2)^{n_2} \dots (g_k)^{n_k} \right\} \left\{ \frac{N!}{n_1! n_2! \dots n_k!} \right\} \right]$$

Hence

$$P = \left[N! \prod_{i=1}^k \frac{(g_i)^{n_i}}{n_i!} \right] \quad \text{--- (4)}$$

Taking logarithm of both sides of eqⁿ (4) we have

$$\ln P = [\ln N! + \sum_i \{ n_i \ln(g_i) - \ln(n_i!) \}] \quad \text{--- (5)}$$

According to Stirling's approximation

$$\ln N! = (N \ln N - N)$$

So that eqⁿ (5) becomes

$$\ln P = N \ln N - N + \sum_i \{ n_i \ln(g_i) - n_i \ln(n_i) \}$$

$$\therefore \ln P = N \ln N + \sum_i \{ n_i \ln(g_i) - n_i \ln(n_i) \} \quad \text{--- (6)}$$

$$[\because N = \sum_i n_i]$$



Differentiating eqⁿ (6) we have:

$$\begin{aligned} \delta(\ln P) &= 0 + \sum_i \left\{ \ln(g_i) \cdot \delta n_i - \ln(n_i) \delta n_i - n_i \cdot \frac{1}{n_i} \delta n_i \right\} \\ &= \sum_i \left\{ \ln(g_i) \cdot \delta n_i - \ln(n_i) \cdot \delta n_i \right\} \\ \text{i.e. } \delta(\ln P) &= \left[- \sum_i \left\{ \ln(n_i) - \ln(g_i) \right\} \delta n_i \right] \quad \because \sum_i \delta n_i = 0 \quad \text{--- (7)} \end{aligned}$$

Now,

The no. of particles is so large that any small change in this no. is not going to materially affect the distribution.

$$\text{i.e. } \delta P = 0$$

$$\therefore \delta(\ln P) = 0 \quad \text{--- (8)}$$

from eqⁿ (7) and (8) we have

$$\sum_i (\ln n_i - \ln g_i) \delta n_i = 0 \quad \text{--- (9)}$$

To eqⁿ (9) we add the auxiliary condition of

(i) conservation of particles $-\sum_i \delta n_i = 0$ --- (a)

(ii) conservation of total energy $-\sum_i U_i \delta n_i = 0$ --- (b)

Multiplying eqⁿ (a) by α and (b) by β and adding to eqⁿ (9) we obtain

$$\sum_i \left\{ (\ln n_i - \ln g_i) + \alpha + \beta U_i \right\} \delta n_i = 0.$$



Thus,
$$\sum_i (\ln n_i - \ln g_i) + \lambda + \beta u_i = 0 \quad \text{--- (10)}$$

for eqⁿ (10) to hold, the quantity in parenthesis must be valid for each value of i .

$$\therefore \ln n_i - \ln g_i + \lambda + \beta u_i = 0$$

$$\therefore \ln \left(\frac{n_i}{g_i} \right) = -\lambda - \beta u_i$$

$$\text{or } \frac{n_i}{g_i} = e^{-(\lambda + \beta u_i)}$$

$$\text{Hence, } n_i = g_i e^{-(\lambda + \beta u_i)}$$

$$\therefore \boxed{n_i = \left\{ \frac{g_i}{e^{(\lambda + \beta u_i)}} \right\}} \quad \text{--- (11)}$$

Eqⁿ (11) is known as Maxwell-Boltzmann distribution Law or M-B statistics